

Interdisciplinary Group project Proposal

Sentiment Analysis and Suspicious Review Detection on Amazon App Review

Team 4

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Sentiment Analysis and Suspicious Review Detection on Amazon App Review

Introduction

Reviews serve as a major source of insight for users, buyers, app developers, marketing managers which influences app download decisions, product buying decisions, ranking, improvements. Reviews are basically electronic word of the mouth but accurately determining if a review truly reflects the product can be quite challenging. That's where sentiment analysis and fake reviews detection comes in, to get the overall sentiment of a review and detect malicious or deceptive reviews aimed at spoiling brand reputation, misleading buyers and user including marketing managers and app developers. Reviews are often very short, informal, generic, misleading, overly exaggerated positively or negatively. To address these issues, this project proposes the use of machine learning based system to analyze and predicts sentiments including fake review detection on amazon reviews. This project is supported by a research-driven understanding of existing limitations and enhanced by the development of a web application that demonstrates results interactively.

Initial Research (State of the Art)

Understanding sentiment and detecting fake reviews have been widely explored in the field of Natural Language Processing. Traditional lexicon-based tool like VADER (Hutto and Gilbert, 2014) is commonly used due to their simplicity, but research has shown key limitations.

Limitations VADER Sentiment Analysis

Hutto and Gilbert (2014) highlight that the model performs better on social media texts but struggles with exaggerated expressions and slangs related to some specific domains. VADER is Rule based; it cannot interpret texts filled with sarcasm, hyperbole or deceptive text written intentionally.

Furthermore, (Ribeiro *et al.*, 2016) further justifies the above notions, the model underperforms on some vocabulary related to the products being reviewed, shows in accuracy in noisy and informal text.

Sentiments analysis of deceptive text revealed that fake or manipulative reviews often use exaggerated positive language, Rule based tools like VADER misclassify these deceptive reviews because they rely on some dictionary-based word sentiment like (good, happy etc) rather than patterns in the context of the review (Fornaciari and Poesio, 2012).

Deceptive reviews could use overly enthusiastic or generic praise, which causes rule-based tool to incorrectly label these reviews as strongly positive. (Ott *et al.*, 2011). This reveals a critical gap in lexicon-based sentiment analysis as it not only misinterprets emotional tone it also fails to detect deception in the written text.

Reviews can be multilingual or non-English, applying a tool like VADER (or any English-only sentiment lexicon) can be less accurate because translated text may lose the original tone, leading to misinterpretation of emotions.

Fake Review Detection

Studies have shown that Fake reviews detection is used complementary with the use of both rule-based heuristic and machine-learning models. Using only rule-based heuristics is not effective enough especially in modern times with the use of different technologies to bypass these rules.

Rule-based heuristics

Datasets often lack explicit truth labels; researchers have sought alternative methods, such as behavioural or structural cues, to indicate suspicious activities. Examples include unusual username format, generic praise, low engagement on reviews (e.g., thumb score, likes, or helpfulness), mismatch between star reviews and text, and review count/length (Mukherjee *et al.*, 2013).

Machine Learning models

In recent time, moving away from the traditional approach, recent research has focused on supervised learning methods such as logistic Regression, Naives Bayes, Support Vector Machines, Random Forests due to their ability to capture contextual meaning, translated texts, domain specific words (Wang and Manning, 2012). Additionally, advance research makes use of transformer models like BERT (Devlin *et al.*, 2019). Models like this, requires large amounts of labelled data and significant computational resources, which may not be practical for all applications.

Furthermore, transformer model can produce results that are difficult to interpret sometimes, potentially distorting explainability in crucial applications such as fake review detection. Machine learning model uses features like (TF-IDF, n-grams, embeddings) and meta data features like review length, timestamps, engagement metrics to detect deceptive reviews. (Ott et al., 2011) Many studies have concluded that effective fake review detection combines the use of textual and behavioural cues as deception can manifest in both linguistic style and behavioural patterns.

A major challenge when using machine learning in fake reviews detection is lack of labelled datasets. Several researchers have proposed the use of “weak labels” based on the patterns of the ratings such as exaggerated positivity, unusual word length (Fornaciari & Poesio, 2012). However, weak supervision introduces its own challenge as model trained on imperfect or noisy labels may learn spurious correlation or fail to generalize well to genuine cases. while weak supervision enables machine learning models to learn patterns of fake reviews even without human-verified labels, it is often necessary when working with large datasets, careful validation and critical assessment of model outputs remain essential.

Explainability

The use of explainability is essential, as it provides insight and answers the 'why' question. Why a model produces a particular output. There may be cases where models mislead or unfairly flag genuine reviews, so showing a transparent process behind the reasoning is necessary for trust and ethical concerns (Guidotti et al., 2018).

LIME (local interpretable model-agnostic explanations) (Ribeiro *et al.*, 2016) and SHAP (Shapley Additive explanations) (Lundberg, Allen and Lee, 2017) provides highlight on which words contributed most to the prediction, allowing users to see whether exaggerated positivity, generic reviews, or unusual review patterns, review length, influenced the model. This is especially important when reviews are translated from other languages, where sarcasm, tone, or cultural meaning can be lost during translation.

Financial Impact of Fake Reviews

The fake review system at Amazon creates a deep market problem which affects more than just customer trust. The fake review problem affects 30% of online reviews (Vazquez, 2023) which results in \$3.8 trillion worth of e-commerce transactions (World Economic Forum, 2022). Amazon removed 275 million suspected fake reviews from its platform during 2024 (Amazon, 2024) yet the issue continues to generate financial damage throughout the entire e-commerce supply chain affecting buyers and authentic sellers and delivery services and the platform operations. This gives insights on how fake reviews impact financial performance of stakeholders and generates monetary losses.

Consumers

Globally, fake reviews cost consumers an estimated \$0.12 of every dollar spent online, leading to a projected loss of over \$770 billion by 2025 (Capital One Shopping, 2025). These losses are because of purchasing poor-quality products, increased return costs, and reduced choice. In the US, households lose about \$2,385 annually in fake review-influenced decisions (The Transparency Company, 2024). The long-term cost includes lack of trust in online markets, reduced purchase efficiency, and lower overall welfare.

Small Sellers

Honest Amazon sellers lose revenue to competitors who use 5-star reviews to climb the rankings. The fake reviews are causing an estimated \$300 billion in annual consumer harm in service sectors, including home repair, legal, and medical services (The Transparency Company, 2024). Fake negative reviews can cut sales by up to 25% when consumers suspect inauthentic feedback (McKinsey, cited in Reputational Sciences, 2025). A US analysis estimated \$300 billion in lost revenue in sectors due to review fraud (Chopra and Evans, 2023). Sellers also face pressure to resort to fraudulent tactics, which can lead to high reputational defense costs. Account bans due to fake review links can lead to total revenue collapse for some small vendors (Amazon, 2024)

Amazon

Amazon reportedly spent \$500 million annually combating fake reviews, employing over 8000 staff for enforcement (Capital One Shopping, 2025). Legal risk grows in US and EU laws, with possible fines up to 10% of global turnover under the UK's Digital Markets, Competition and

Consumers Act 2024 (UK Government, 2024). If customers' trust falters, 2% drop in engagement could cost Amazon \$12 billion in total annual cost.

Advertiser

Fake reviews reduce ad efficiency. Sellers face ad costs competing with boosted listings. Misaligned spending leads to low ROI and budget cost. Globally, \$152 billion misdirection in consumer spending is caused by fake reviews, which advertisers compensate for (World Economic Forum, 2022).

Brands

Brands lose market share to unorganic review and suffer damage from malicious one star reviews. Each fake review reduces brand equity, increases costs of refund, and misguides R&D. United Kingdom authorities noted billions in brand losses from fake reviews-driven diversions (CMA, 2022). Some companies, like Nike, pulled their products from Amazon entirely due to trust concerns.

Delivery & Fulfillment Services

Misleading reviews increase return rate and inflation cost. In the US \$743 billion in returns logged in 2023, a portion of which stemmed from falsely rated products (NRF, 2024). Each return imposes transport, labor, and packaging costs, reducing logistics margins and increasing carbon footprint.

Regulators

Regulatory bodies globally bear the cost of investigating and monitoring fake review. The FTC's news rules 2024 allow civil penalties of up to \$51,74 per violation (FTC, 2024). These funds were redirected from proactive initiatives to fraud policing.

Digital Economy

Fake reviews compromise AI algorithms, reduce innovation incentives, and distort spending patterns. Trust loss reduces the growth of e-commerce and review-based platforms, showing digital GDP expansion and reducing market efficiency.

Problem Statement

Digital platforms rely on user reviews which helps guide product decision making and build consumer trust. However, interpreting these reviews accurately can be challenging because the text is often unstructured, subjective, inconsistent, and sometimes intentionally exaggerated. These reviews tend to express mixed or unclear sentiment, written in informal language, includes misleading or low quality information. As a result, developers and organisations find it difficult to discover reliable insights into user experience and prioritise meaningful improvements. Additionally, most publicly available datasets lack explicit and authentic labels, making identification of suspicious or potentially deceptive reviews more difficult.

Therefore, the key problem this project addresses is the difficulty of accurately analysing the sentiments of reviews generated by user and detecting potentially deceptive or low quality reviews. To solve this, the project will focus on developing a reliable and transparent system that can analyse reviews, predict sentiment, and flag suspicious patterns using machine-learning techniques. Metrics of evaluation for assessing the model's performance will include accuracy, precision, and its F1 score on sentiment analysis and suspicious review detection tasks. The success of the project will also be measured by the usability and interpretability of the web application, ensuring that explanation outputs increase user trust. Integrating explainability into the web application further ensures there is transparency, fairness, and users' confidence in the system's outputs

Project Objectives & Scope

1. The project aims to develop and evaluate a machine-learning sentiment model using TF-IDF features by the midway of the project timeline.
2. Identify and test a machine learning method for suspicious review detection using both text and metadata features within the first half of the project.
3. Extract and analyze linguistic (e.g., text length, generic phrases) and behavioural features (e.g., username patterns) that will be used as inputs for the generation of labels for the suspicious review model.
4. Develop a Streamlit web application that displays sentiment predictions, suspicious-review flags, and explanation outputs by the final development sprint.
5. Integrate the use of LIME or SHAP explanations into the web application to increase model transparency and user confidence before the testing phase begins.
6. Document a full project report summarizing the methods, issues encountered, solutions, results, limitations, and ethical considerations including recommendations for future work before submission week

In Scope:

1. Development of a machine learning sentiment analysis model.
2. Exploration of suspicious review detection strategies using text and metadata features.
3. Development of a Streamlit web application to demonstrate model predictions.
4. Integration of explainability techniques (e.g., LIME/SHAP) to ensure transparency.
5. Foundational literature review to inform model design and justify approach.

Out of Scope:

1. Deep-learning or large production-scale models, such as BERT or neural networks requiring heavy computation.
2. Mobile application development, including Android/iOS deployment
3. Commercial deployment or enterprise level optimization

Work Packages

Given the nature of the tasks, the project is anticipated to progress through several iterations. After each iteration, outcomes will be critically reviewed by the team, and adjustments to models or processes will be implemented based on observed results and newly identified requirements.

WP1: Learning and understanding of relevant tools/models and data sets

1. Python, Scikit-learn, Streamlit, version control (GitLab)
2. Data set exploration

WP2: Data Understanding and Preparation

1. Import the data set
2. Load and Amazon review data set
3. Explore metadata structure such as usernames, ratings, timestamps, etc.

WP3: Exploratory Data Analysis

1. Descriptive analysis:
compute summary statistics: missing values, multiple language, reviews length, etc.
2. Visualization: Plot charts and word clouds to understand patterns and other features in the data set

WP4: Data Cleaning & Preprocessing

1. Cleaning and normalization: Lowercasing texts, punctuation removal, removal of stop words, tokenization, and handling multilingual reviews.

WP5: Text Representation

1. Convert cleaned text into numerical features using tools such as TF-IDF vectorization to support the machine learning model's understanding.

WP6: ML Sentiment Model Development

1. Choose the machine learning method.
2. Build the machine learning model.
3. Train the sentiment models.
4. Test the model accuracy.

WP7: Suspicious Review Detection Exploration

1. Identify a suspicious pattern in the review to label the data set

2. Train the model based on the labels.
3. Provide basic predictions

WP8: Explainability Development

1. Application of explainability methods: Tools such as LIME and SHAP to interpret model predictions

WP9: Web App Development

1. Streamlit prototype implementation
2. Design a user interface for entering reviews, displaying sentiment, flagging fake reviews, and providing explanations.
3. Integrate the models into the Streamlit front end for real-time use.

WP10: Testing, Documentation, Reporting

1. test the full pipeline from end to end with different reviews
2. document and reflect on the tasks
3. Prepare final report and presentation.

Proposed methods for project management and control

We will use Agile project management for this project, with GitLab supporting collaboration, version control, and workflow organization. Within the team, specific roles are assigned to optimize efficiency: one member will focus on data preprocessing and cleaning, another will handle feature engineering and implementation of machine learning models, and a third will develop the web application interface and integrate explainability methods. All team members will contribute to documentation and testing.

Agile is a good fit for machine learning projects because development is usually iterative. Saltz and Shamshurin (2016) point out that data science and ML projects rarely move in a straight line, so teams need to reassess and adapt often. With Agile's sprint-based approach, we can incorporate new insights from Exploratory Data Analysis (EDA) and model testing into each cycle, helping the project continue to improve.

We will use GitLab as our main platform to track progress, assign tasks, and organize our code. GitLab provides tools such as merge requests, branch management, issue tracking, and Kanban boards that support teamwork in ML projects (Amershi et al., 2019). Since each team member will focus on different tasks, GitLab helps us keep everything organized and avoid conflicts. Its documentation features also make our work transparent, which is important for group projects that will be assessed.

We considered other project management methods, but Agile is the best fit for this project. The Waterfall model is structured and easy to document, but it assumes requirements will remain the

same and that little revision will be needed. However, ML workflows often require reverting to earlier steps, such as changing preprocessing or retraining models, which Waterfall does not handle well (Royce, 1987).

Microsoft teams will be used for standups meeting to check in on team members progress, this helps tackle difficulties as the project progresses.

Dataset Availability & Licensing

The project uses a publicly available Amazon review dataset from Kaggle. The dataset is licensed under **CC-BY-NC-SA 4.0** (Creative Commons Attribution–NonCommercial–ShareAlike 4.0), which means we are allowed to share and adapt (modify, transform, build) the dataset for research and academic purposes, provided the source is credited, and the data is not used for commercial purposes. The dataset is uploaded by Ashish Kumar who is the dataset creator and curator.

The data set was initially collected on the 12th of September 2018, and it is still currently updated daily, which the dataset reflects ongoing, real time user activities making it more representative of current e-commerce review patterns. The `amazon_reviews.csv` file contains: 81,705 reviews as at the last download (9th December 2025), multiple columns including review ID, username, score, thumbsupcount, review created version, at (timestamp) and app version. The data set is large enough to support machine learning modelling.

Research Gap

We have reviewed other work done with this data set and most of these works focuses on basic sentiment analysis using, VADER, Text Blob, simple machine learning models, but they rarely address suspicious review detection. Existing work on this dataset is typically limited to polarity scores and visualizations without including metadata features, explainability or real time interactive web app.

This project differs by combining text-based sentiment modelling with suspicious review detection using language and behavioural cues such as review length, rating and review mismatch, username patterns. Additionally, the explainability feature using LIME/SHAP will be integrated, and a web app will be developed to make results transparent and interactive.

This ensures that the project contribute a unique and research driven approach beyond what has been done on Kaggle.

Ethical Considerations

Ethical considerations include:

1. user privacy: Information only username, no sensitive data is shared.
2. There could be potential bias in the machine learning prediction.
3. Transparency: Explanation of suspicious Review output will be included in the results.
4. Respecting the data set licensing.

5. Flagged reviews are seen as indicators, not final validation of scam activities, to avoid misuse of the results derived.

Risk Management

A risk register was created to identify potential risk, assess the severity and provide proper risk mitigation action to support the successful project delivery.

The risk register is provided in Appendix 3

Project Timeline

The project follows an agile sprint structure, and an overall Gantt chart was created to visually map the estimated timeline of the major task and milestones throughout the project. The Gantt chart serves to align project sprints with key deliverables, providing a high-level overview that helps the team coordinate iterative cycles with scheduled deadlines.

The full Gantt chart can be found in Appendix 1

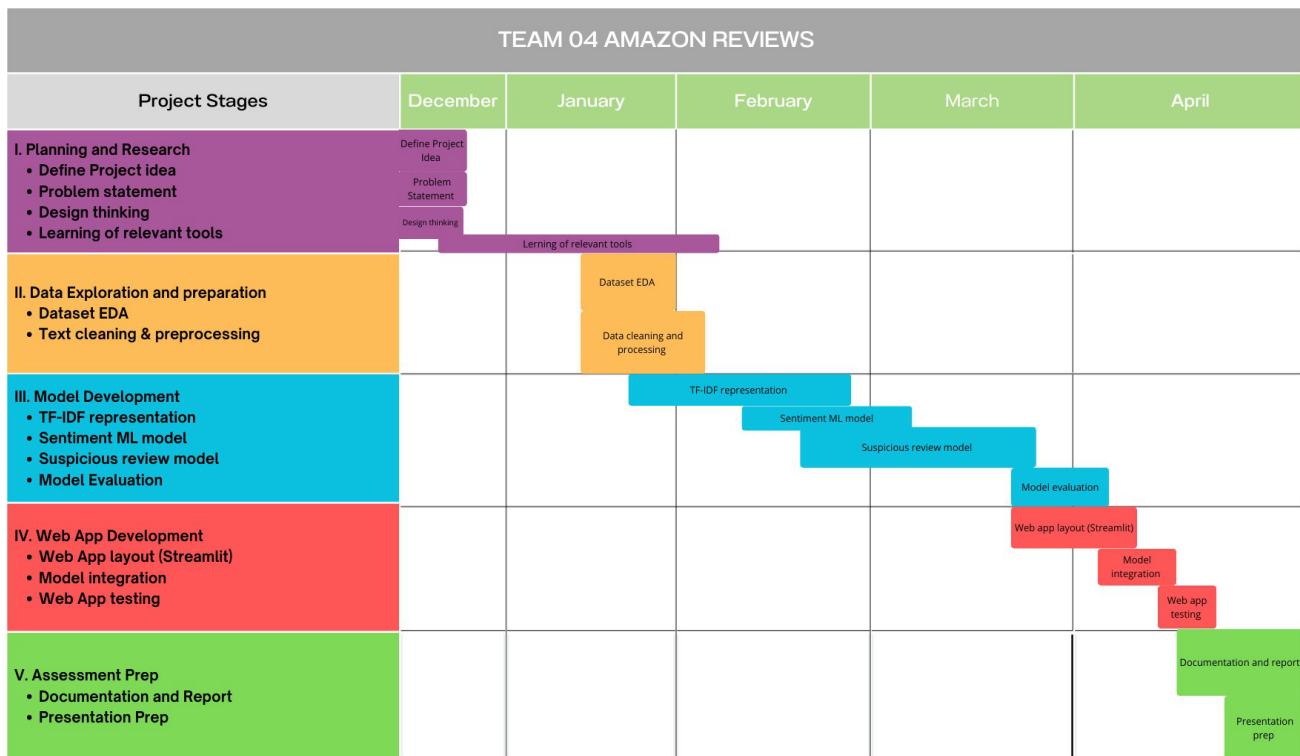
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Appendix

Appendix 1: General Timeline of delivery



Appendix 2: Possible Web Application Features

Features	Description
Input Review box	Text field where users type or paste any app review.
Sentiment prediction	Display whether the review is positive or Negative
Suspicious Review indicator	Flag a review as Genuine or potentially suspicious
Explainability	Show key words or features that influenced the model's decision
Real time result	Provide output as soon as review is pasted
Clean streamlit interface	Simple layout, buttons, color coded results

Appendix 3: Risk Register

ID	Description of Risk	Likelihood	Impact	Severity	Risk Mitigation	Risk Owner
1	Dataset contains missing, noisy, or multilingual reviews	Medium	High	High	Clean dataset, handle multilingual text	Data Lead
2	Model underperforms due to limited or unbalanced data	Medium	High	High	Improve preprocessing, tune model	Model developer
3	Web App fails to load ML models during integration	Low	High	High	Test model loading separately, consistent paths	Web App developer
4	Limited knowledge of key tools slows development	High	Medium	Medium	Allocate learning sprints, share resources	Team Members
5	Team communication challenges causing delays	Medium	High	High	Weekly standups, GitLab issues, WhatsApp	Scrum Master
6	Loss of progress from Git errors or missing files	Low	High	Medium	Regular commits, branch strategy, backups	Team members
7	Model incorrectly flags genuine reviews	Medium	Medium	Medium	Use LIME/SHAP explainability	ML Team
8	Inability to meet project Timeline	Medium	High	High	Break tasks into smaller units, use a short agile sprint, do weekly check-ins	Scrum Master

